

Temporal and Spatial Distortion for VR Rhythmic Skill Training

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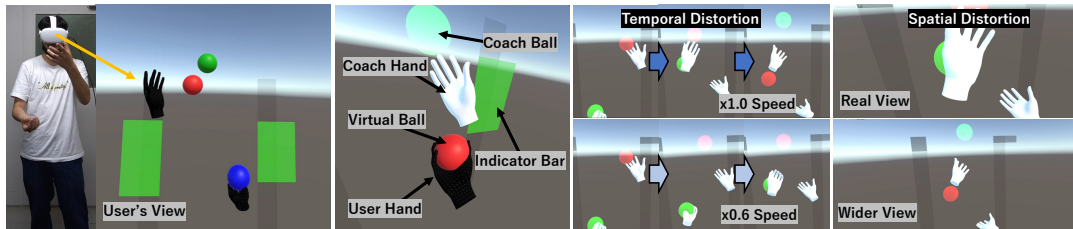


Figure 1: The overview of the VR Juggling System, Visualization Functions, Temporal Distortion, Spatial Distortion.

ABSTRACT

In many sports, rhythmic skills are considered important. In this paper, we take juggling as an example and propose a VR system that simplifies the acquisition of a sense of rhythm. The proposed system uses temporal and spatial distortion and other functions to assist the training. A pilot study is conducted to validate the effectiveness of each function.

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1 INTRODUCTION

Rhythmic skills such as dancing, skiing, and juggling require good coordination of body postures and a sense of rhythm. Much existing skill acquisition research only focuses on how to simplify the learning of mimicking an optimal posture by providing error feedback or reducing the speed of the motion. However, these methods are only effective for posture understanding, which might interrupt the continuous motion and affect the learning of a rhythmical motion. In this paper, we focus on simplifying the acquisition of a sense of rhythm for rhythmic skills.

Therefore, we take ball-juggling as a good example where a sense of rhythm is considered to be one of the most essential factors. When learning juggling, the major difficulties of the rhythmic motion are the fast rhythm for the hand to get used to and the many information for the eyes to catch. To solve these problems, we develop a virtual reality (VR) training system using temporal and spatial distortion. The users can perform realistic juggling in

VR while the temporal speed and spatial field of view are gradually modified according to their performances. An initial pilot study with a comprehensive interview is conducted to prove the concept and evaluate the effect of each function in a prototype system.

2 RELATED WORK

In terms of juggling training, Borglund et al. created a VR juggling system that provides instructions on ideal ball height and timing of throws [Borglund et al. 2021]. However, previous juggling works mainly focus on simulating real juggling without making use of the virtual environment.

For using physical distortion in virtual training, Matsumoto et al. created a system to change the speed of time in VR skiing to improve the training efficiency [Matsumoto et al. 2022]. As a distortion to space rather than time, Higuchi et al. created a system that uses a flying robot that follows the user to experience a different field of view than the actual one [Higuchi and Rekimoto 2013].

3 SYSTEM DESIGN

As shown in the Figure 1, we implement a VR juggling system. Different from previous works, we use the hand tracking module of the Quest V2 HMD thus users do not need to hold the controller, allowing for highly immersive training. For the visual information, we refer to previous VR systems that display users' hands and the coach's ideal hand movement and two indicator bars show the traveled distance of a throw.

3.1 Assist Functions

To support the acquisition of rhythmic sense, the following three functions are implemented.

Audio Assist. As a basic audio cue for instruction of the rhythm, the timings when the coach throws and catches the balls are presented by sonification.

Temporal Distortion. In the real world, balls must be thrown at least once every 0.5 seconds for a three-ball-juggling, which is difficult for beginners to get used to. Therefore, a temporal distortion is

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performed to the virtual temporal space, where the speed of time can be adjusted according to user's skill level. Beginners can start from a slow speed, which makes it easier to understand the motion.

Spatial Distortion. Another problem is the great number of factors to focus in juggling, such as the movement of hands and the trajectory of balls, which makes it difficult to concentrate on the rhythm. That is why a beginner frequently move their head while an experienced juggler can fix their eyesight. Therefore, in addition to the temporal distortion, we also distorted user's position by moving them backwards than their real position, which expand the field of view of the users and make it easier to understand the overall movement.

4 PILOT STUDY

A pilot study was conducted to evaluate the system and each function. The participants were six beginners (two females, four males, Mean age=23.3, SD=2.05), all of whom had some VR experience.

First, the participants watched a video showing how to practice juggling and trained with an actual ball for about five minutes. After the first practice, the participants tried the three-ball flash (holding three balls in each hand, throwing and catching all balls once) 10 times to measure their base juggling skills. After that, we conducted a 15-minute VR training using the proposed system, in which the users experienced various assist functions, the users were asked to try the three-ball flash 10 times again using a real ball. Finally, we interviewed the participants and obtained feedback on the usefulness of each assist function for training, their impression of the system, and suggestions for improvement.

4.1 Skill Test

The results of the juggling skill test before and after the VR training are shown in the Figure 2. The number of misses decreased and the number of near misses, in which the participants were able to catch two out of the three balls, increased, suggesting that the training had a certain effect. A participant was able to continue juggling beyond the three-ball flash after the VR training.

4.2 Interview

Many participants commented that the Coach Hand, the Indicator Bar and the Audio Assist could be used effectively when training with two balls, but when training with three balls, the amount of information provided was too much and became a distraction. For the two different distortions in time and space, most of the participants had positive comments that both were useful for training. In the skill test after the VR training, some participants commented they were able to move with an ideal rhythm in mind. There were some comments that the current VR system does not provide the kind of training that prevents the failure to throw the real ball forward.

5 DISCUSSION

As a result of the pilot test and participants' feedback, several insights for improving the system are obtained.

First, in an undistorted normal condition, the audio and visual instructions seem to be less helpful because of the great number of

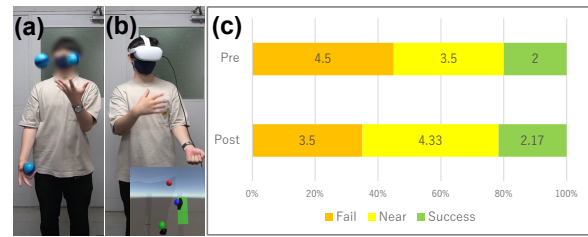


Figure 2: Real Training(a), VR Training(b), The performance before and after training in VR(c). Success: Catching all three balls, Near: Catching two out of three balls, Fail: Others.

information. When using the proposed distortion method, most participants feel easier to follow the instruction and catch the rhythm. Actually, the number of misses decreased in the skill test after the VR training, and some participants commented that they became more aware of the ideal rhythm, which suggests that the proposed system improves their short-term sense of rhythm.

Besides the time distortion function, which has already proved to be effective for motion mimicking in other sports, a new finding is the effect of the spatial view distortion. Although there is a problem with the viewpoint being moved backward, causing the virtual hands to differ from their actual position, it is thought that the increased field of view is more effective than this.

For future work, we plan to create a system that analyzes the user's skills and error tendencies and automatically presents only the necessary elements. This allows effective assistance while keeping the amount of information tiny. In the current implementation, the speed magnification and field of view are modified manually, but this could also be changed automatically based on the success of the user's level in the future. Last but not least, we plan to consider training systems that combine temporal and spatial distortion for more complicated rhythmic skills or sports.

6 CONCLUSION

In this paper, we proposed a VR training system that simplifies the acquisition of a sense of rhythm. Feedback from users indicated that not only temporal distortion but also spatial distortion was effective, suggesting future training applications.

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